

Advanced (Habanero)

- Q1.** A bakery sells x bags of bread and y cakes. The profit is $2x + 3y$ dollars. If $x + y \leq 100$ and $2x + y \leq 150$, what is the maximum profit?
- (A) 200
(B) 300
(C) 350
(D) 400
(E) 450
- Q2.** A farmer has x acres of wheat and y acres of corn. The total area is $x + y \leq 200$ acres. If $x \geq 50$ and $y \geq 50$, what is the minimum number of acres of wheat?
- (A) 50
(B) 75
(C) 100
(D) 125
(E) 150
- Q3.** A company produces x units of Product A and y units of Product B. The profit is $3x + 2y$ dollars. If $x \geq 0$, $y \geq 0$, and $x + y \leq 100$, what is the maximum profit?
- (A) 200
(B) 300
(C) 500
(D) 600
(E) 700
- Q4.** A city has x buses and y trains. The total number of vehicles is $x + y \leq 50$. If $x \geq 10$ and $y \geq 10$, what is the maximum number of buses?
- (A) 20
(B) 25
(C) 30
(D) 35
(E) 40
- Q5.** A student has x hours for homework and y hours for study. The total time is $x + y \leq 10$ hours. If $x \geq 2$ and $y \geq 2$, what is the maximum number of hours for homework?
- (A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q6.** A store has x boxes of pens and y boxes of pencils. The total number of boxes is $x + y \leq 20$. If $x \geq 5$ and $y \geq 5$, what is the maximum number of boxes of pens?
- (A) 10
(B) 11
(C) 12
(D) 13
(E) 15
- Q7.** A manufacturer produces x units of Product A and y units of Product B. The profit is $2x + 4y$ dollars. If $x \geq 0$, $y \geq 0$, and $x + y \leq 50$, what is the maximum profit?
- (A) 100
(B) 150
(C) 200
(D) 250
(E) 300
- Q8.** A factory has x machines for production and y machines for packaging. The total number of machines is $x + y \leq 30$. If $x \geq 5$ and $y \geq 5$, what is the maximum number of machines for production?
- (A) 15
(B) 18
(C) 20
(D) 22
(E) 25

Open Ended Questions (FRQ)

- Q9.** A company produces x units of Product A and y units of Product B. The profit is $3x + 2y$ dollars. If $x \geq 0$, $y \geq 0$, $x + y \leq 100$, and $2x + y \leq 150$, find the maximum profit and the values of x and y that give this profit.
- Q10.** A farmer has x acres of wheat and y acres of corn. The total area is $x + y \leq 200$ acres. If $x \geq 50$ and $y \geq 50$, find the minimum number of acres of wheat and the corresponding value of y .

Chef's Tips

- Read the questions carefully
- Use the given equations and constraints to solve the problems
- Check your work and ensure that your answers are reasonable